Nonconvexity of non-completable probability tables

A counterexample to MathDB 400968

Principia Math in Progress

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Task 1. Resolve the original convexity question

The problem in Kummer's Oberwolfach report [1, p. 1271] asks whether the nonnegative partial tensors which have no completion as the joint distribution of independent random variables always form a convex set. The stated setting has a full-dimensional completion image and $|E| = \sum_i (d_i - 1)$ observed coordinates.

Theorem 1. *The answer is negative, already for a 3×3 table with*

$$E = \{(1,1), (1,2), (2,1), (3,3)\}.$$

This pattern observes every row and every column, and its completion image has nonempty interior in $\mathbb{R}^4$.

Proof. Use the coordinate order $(11, 12, 21, 33)$, and define

$$P = \left(\frac{1}{32}, \frac{1}{8}, \frac{1}{8}, \frac{1}{16}\right), \qquad Q = \left(\frac{15}{32}, \frac{1}{8}, \frac{1}{8}, \frac{1}{16}\right).$$

All entries are positive, and their respective observed sums are $11/32$ and $25/32$, both less than one.

We first obtain a necessary condition for completion. Write a putative completed table as $T_{ij} = a_i b_j$, where a, b are nonnegative probability vectors. If its observed coordinates are (x, y, z, w) with $x > 0$, rank one forces $T_{22} = yz/x$. Consequently

$$S := x + y + z + \frac{yz}{x} = (a_1 + a_2)(b_1 + b_2), \qquad w = a_3 b_3.$$

Cauchy–Schwarz, applied to the two unit vectors $(\sqrt{1 - a_3}, \sqrt{a_3})$ and $(\sqrt{1 - b_3}, \sqrt{b_3})$, gives

$$\sqrt{S} + \sqrt{w} \le 1.$$

In particular, when $w = 1/16$, every completion must satisfy $S \le 9/16$. For our two points, however,

$$S(P) = \frac{25}{32} > \frac{9}{16}, \qquad S(Q) = \frac{361}{480} > \frac{9}{16}.$$

Thus both P and Q are non-completable.

Their midpoint is completable:

$$\frac{P + Q}{2} = \left(\frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{16}\right), \qquad T = \frac{1}{16}\begin{pmatrix} 4 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/4 \\ 1/4 \end{pmatrix}\begin{pmatrix} 1/2 & 1/4 & 1/4 \end{pmatrix}.$$

The entries of T are positive and sum to one, and its entries in E are exactly the midpoint. This proves nonconvexity.

It remains to check the full-dimensional hypothesis. Parameterize the two probability vectors by $a = (a_1, a_2, 1 - a_1 - a_2)$ and $b = (b_1, b_2, 1 - b_1 - b_2)$. The observed-coordinate map is

$$F(a_1, a_2, b_1, b_2) = \big(a_1b_1, a_1b_2, a_2b_1, (1 - a_1 - a_2)(1 - b_1 - b_2)\big).$$

Its Jacobian determinant is

$$\det DF = a_1b_1(a_1 + a_2 - b_1 - b_2).$$

At the interior parameter point $(a_1, a_2, b_1, b_2) = (1/2, 1/4, 1/3, 1/3)$ it equals $1/72$. The inverse function theorem therefore shows that the image contains an open subset of $\mathbb{R}^4$. Finally, $|E| = 4 = (3 - 1) + (3 - 1)$, as required. □

The necessary completion inequality used above agrees with the general matrix-completion framework of Kubjas and Rosen [2]. The counterexample is verified directly here, without requiring a sufficiency criterion for arbitrary partial tables.

Task 2. Classify the convex matrix patterns

Characterize the observed bipartite forests E with $|E| = d_1 + d_2 - 2$ for which the set of non-completable positive partial tables is convex. Include a criterion stated in terms of the connected components of the observation graph.

Task 3. Determine the smallest counterexample patterns

Determine the smallest table formats and observation counts admitting nonconvexity when every row and column is observed and the completion image has nonempty interior.

Task 4. Describe the convex hull

For the displayed 3×3 pattern, determine the convex hull of the non-completable region inside $\{(x, y, z, w) \geq 0 : x + y + z + w \leq 1\}$, including its boundary faces.

Task 5. Extend the analysis to higher-order tensors

Find structural criteria for convexity or nonconvexity for tensor orders at least three, requiring every coordinate value to occur in the observation pattern and retaining the full-dimensional hypothesis.

References

[1] M. Kummer, Rank One Tensor Completion, in *Algebraic Statistics*, Oberwolfach Reports 14 (2017), 1270–1272, published 2018. doi:10.4171/OWR/2017/20.

[2] K. Kubjas and Z. Rosen, Matrix Completion for the Independence Model, Journal of Algebraic Statistics 8 (2017), 1–21. arXiv:1407.3254.